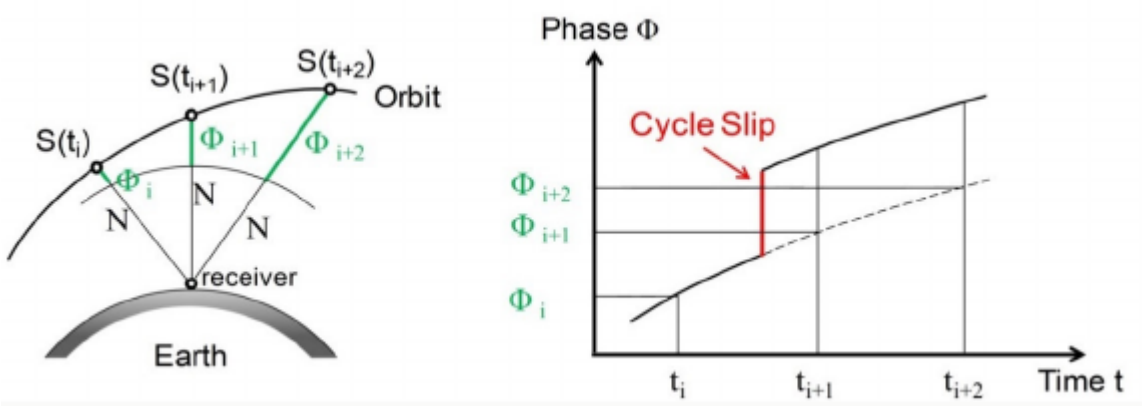
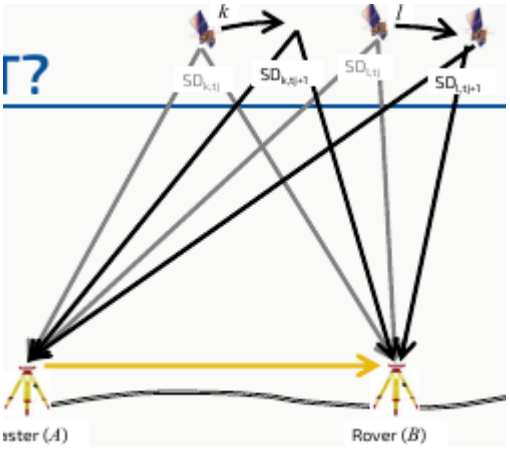
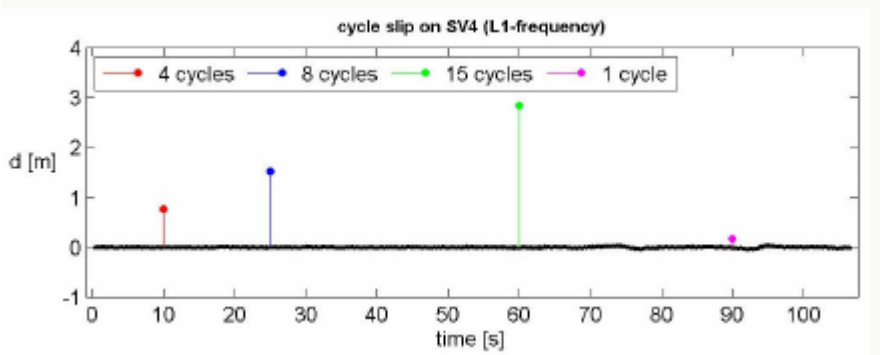
Lecture 7: Observation Methods, Reference Services & Augmentation Systems

- Assumptions so far:
  - Ambiguities do not change over time as long as the satellite is tracked
- But:
  - During tracking, jumps can occur due to bad measurement conditions -> Cycle Slips
  - 

Possible Cycle Slip Detection Methods

- Triple differencing:
  - Compare two consecutive double differences and look for 'jumps'
- Geometry-free linear combination:
  - Create combination of L1 and L2 phase observations, which is independent of the range and look for 'jumps'
- GNSS/IMU integration:
  - Compare with other sources of information such as inertial sensors, and look for inconsistent measurements

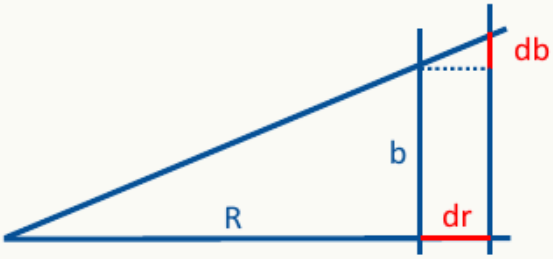




- Assumption so far:
  - The baseline is short (< 10km)
    -> Signals are affected by similar errors
    -> Ionospheric and Tropospheric delays and satellite orbit errors are eliminated by differencing
- But:
  - For long baselines (>10km): errors not fully eliminated
    -> ambiguity fixing becomes more difficult or impossible
- Some measures against:
  - **Ionosphere-free combination**
    - Ionospheric delay is inversely proportional to the square of the frequency
    - **Advantage**:
      - Ionospheric effects are nearly completely eliminated
    - $\varphi_{IF} = \frac{f_1^2}{f_1^2 - f_2^2}\varphi_1 - \frac{f_2^2}{f_1^2 - f_2^2}\varphi_2$
      - **Disadvantage**:
        - Ambiguities are not integer anymore (factors are float values)
        - Noise is higher (~ factor 3)
      - Always assess whether its use is beneficial
  - **Estimate tropospheric wet delay**
    - Tropospheric **conditions** are **different** at two long baseline receivers (such as with large height differences)
    - Dry part can be modelled well, **wet part not**
    - Wet delay estimated as unknown parameter in estimating algorithm
    - Difficult for short or kinematic observations, but always necessary for longer observation durations (>2h)
  - **Use precise orbits from IGS**
    - influence of orbit errors not fully reduced
    - Usage of precise orbit information should be used

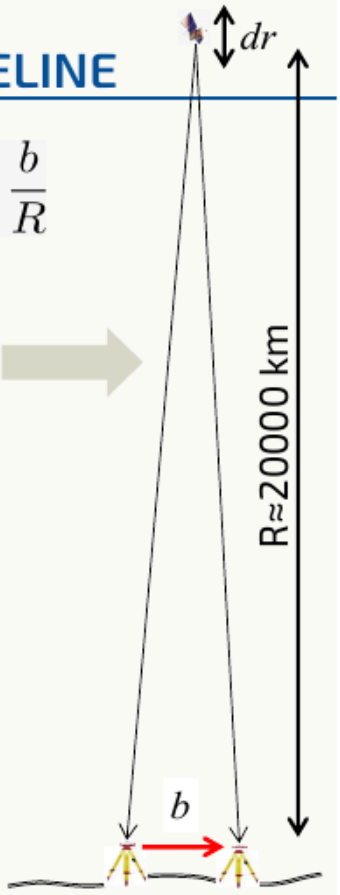
INFLUENCE OF ORBIT ERRORS ON THE BASELINE

- Theorem of intersecting lines (Strahlensatz)



$$\frac{b+db}{R+dr} = \frac{b}{R}$$

$$\boxed{\frac{db}{b} = \frac{dr}{R}}$$



Example: Baseline length 1200km Ignoring all other errors!

- Broadcast orbit data (1m accuracy)

$$db = b \cdot \frac{dr}{R} = 1200km \cdot \frac{1m}{20000km} = 6cm$$

- IGS precise orbits (5cm accuracy)

$$db = b \cdot \frac{dr}{R} = 1200km \cdot \frac{0.05cm}{20000km} = 3mm$$

Method	Concept	Baseline length	duration	Obs.	real time	accuracy
permanent	simultaneous observation of multiple stations	Up to global	Years	Phase	yes/no	mm
static	simultaneous observation of multiple stations with/without change of points	Up to several 100 km	1 h up to multiple days	Phase	yes/no	sub-mm – mm
RTK	Reference station and rover with sequential point measurement	< 10 km	Rover: a few seconds to minutes	Phase and Code	Yes	2-3 cm
PPK		-- as RTK --			No	2-3 cm
DGNSS	Reference station, Rover, Navigation	-	Seconds	Code	Yes	0.5 – 5 m
PPP	Single Point Positioning (SPP) with carrier phase	-	> 30 min	Phase and Code	yes/no	cm-m
SPP	Navigation	-	seconds	Code	yes	5-20 m

relative

absolute

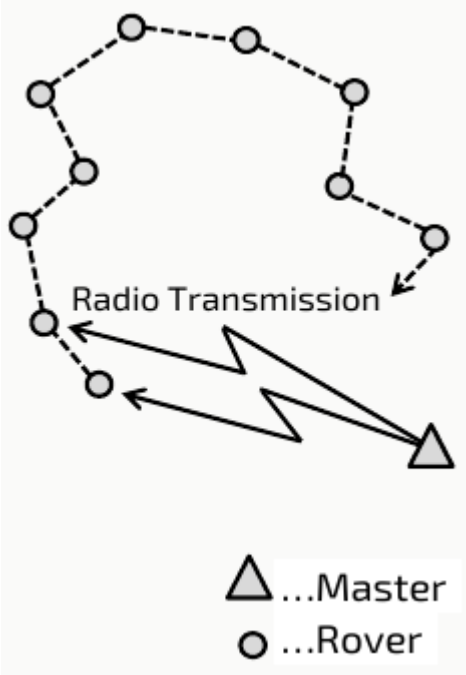
Permanent/static observation methods

- Long observation time (>2h) (reduction of multipath)
- Fixing of ambiguities
- Often simultaneous measurement at multiple stations (networked solution)
- Static GNSS is the usual method for high accuracy measurement
- **Pro: High accuracy**

- **Con: Long observation times**

RTK: Real Time Kinematic (RTK)

- data link between master and rover
- Calculation of the rover position on the rover (coordinates are directly available in the field)
- Standard methods of stake-out and point measurements
- Fast ambiguity fixing necessary (at least 2 Frequency observations)
- Measuring time: 10s - 20 min (or faster)
- No data recording between the points
- Baseline length limited by the radio link



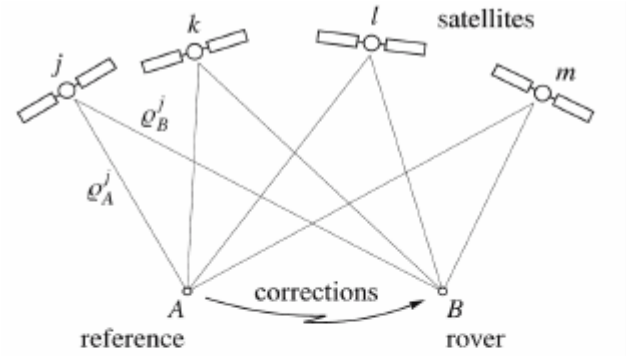
- Transmission of data from a master / reference station is necessary for the real-time method
  - data: observations or correction data
  - radio link: radio, GSM, Internet
- Data from
  - Own reference station
  - Foreign reference station
- Data format: RTCM (standard), sometimes supplier have proprietary formats, needing less bandwidth

SPP

- Navigation solution
- Single receiver, code observations
- At least four satellites visible

- **Pro:** absolute position, no data link necessary
- **Con**: Low accuracy
- -> Enough for coarse navigation in existing maps
- -> Accuracy enhancement possible using SBAS
- -> Suitable for geodetic applications only as intial estimate

DGNSS

- Difference to relative GNSS using carrier phases -> no double differencing of measurements, but applying corrections to pseudoranges
- Coordinates of the reference station are known -> difference to the expected pseudorange is calculated and transmitted to the rover
- Data format: e.g. RTCM
- **Pro:** More accurate than SPP solution
- **Con**: Accuracy depends on baseline length
- 

  Hofmann-Wellenhof (2008), Fig. 6.1

  $$corr = \rho_{A,computed} - \rho_{A,observed}$$
  $$\rho_{B,corr} = \rho_{B,observed} + corr$$

  Not used very often: if reference and rover, RTK is preferred

PPP

- SPP but with carrier phases
  - -> absolute carrier phase based processing
- Accuracy: ~cm
- All observation errors are modelled
- For Real-time processing a connection to a service is necessary (e.g. IGS Real Time Service)
- Also in Post-Processing Webservices are necessary
- **Pro**: High Accuracy with only one receiver

- **Con**: Long initialisation time needed (~30min)

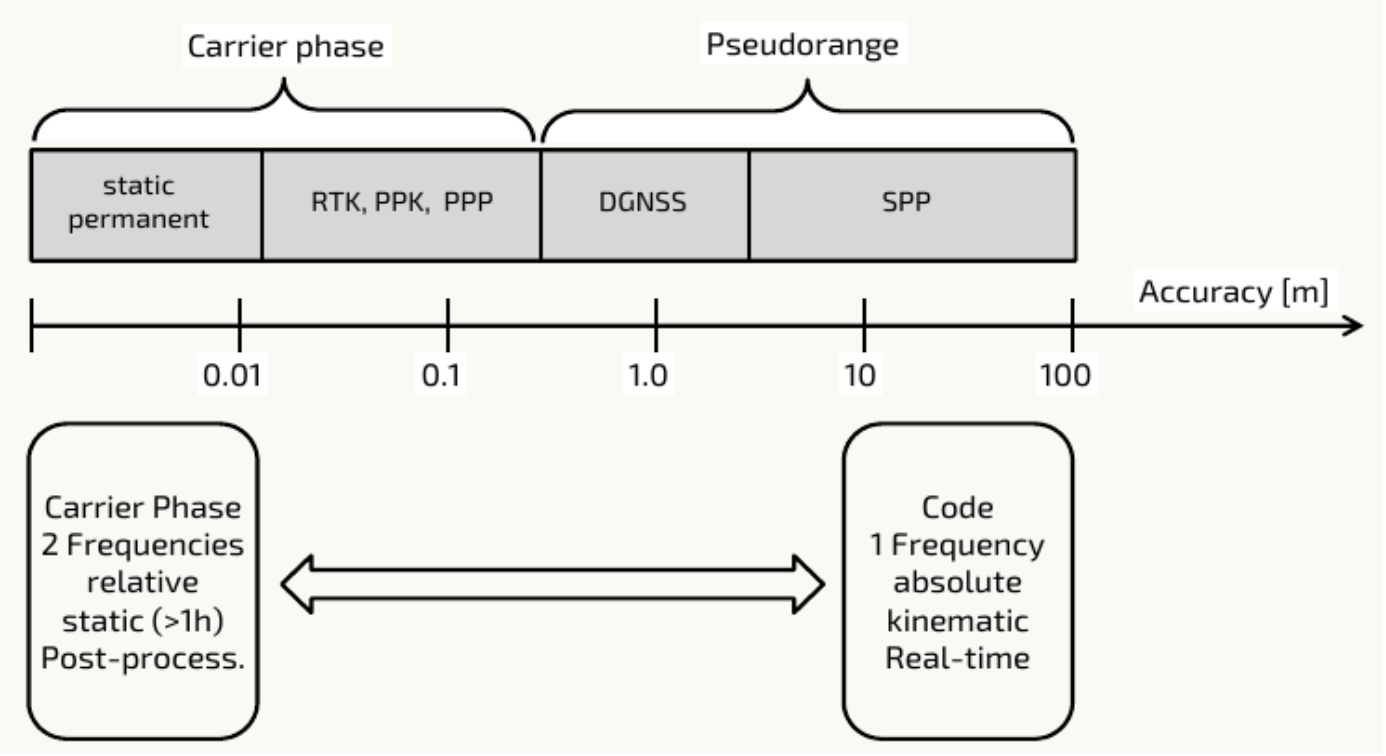
Static Vs. Kinematic Processing

Static Processing

- 3 Position parameters (x, y, z) are estimated for the whole observation time
- Only possible, if the receiver is **not** moving

Kinematic Processing

- For each observation epoch 3 position parameters are estimated
- Usually applied in RTK, PPK, DGNSS and SPP
- Always used for kinematic/moving objects
- Can also be performed for static antennas!



Network RTK

Flaws of RTK/PKK GNSS:

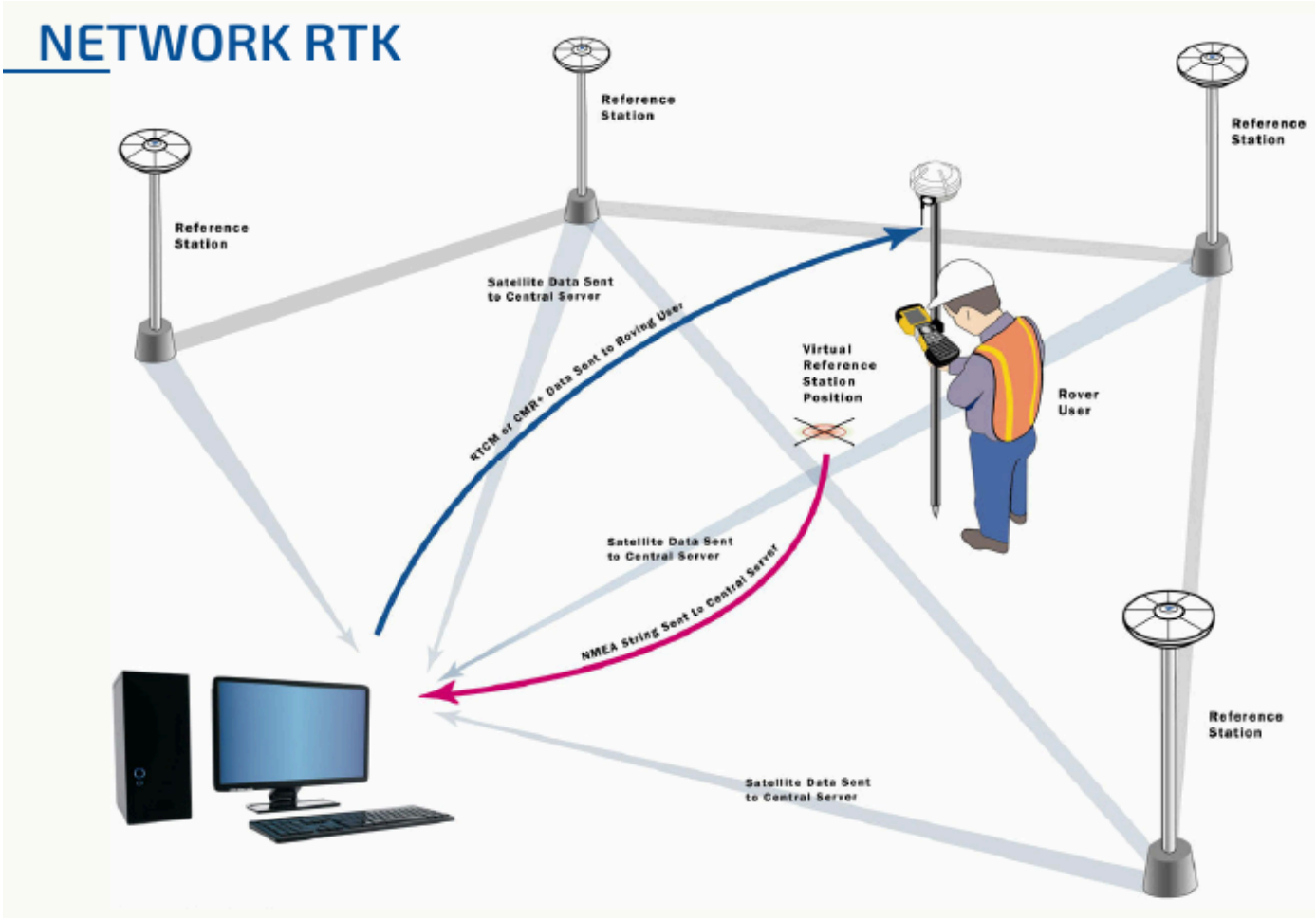
- ~cm accuracy within seconds/minutes or real-time
- Limited baseline length (<10-30km)
- Sometimes not robust (ionospheric disturbances) or wrong (Only uses one reference station)

-> Network RTK

- Network of three or more reference station with well known positions
- Observation errors at the stations can be calculated

- Observation errors can be interpolated over the whole area between the stations
- Information about the errors are transmitted to the rover
- This service is usually done by a reference service provider
- Methods differ in the type of information transmitted to the rover
    - Virtual Reference Station (VRS)
    - Flächenkorrekturparamer (FKP)
    - Master Auxiliary Concept (MAC)

Virtual Reference Station (VRS)



1. Rover calculates a coarse position using SPP and transmits it to the central server
2. Server calculates the observation errors for that position based on all observations of the reference station network
3. Server simulates a reference station at that position (calculates artificial observations for this position)
4. Server sends these observations to the rover
5. Rover calculates a position relative GNSS based on carrier phases using this observations (Short baseline RTK)

E.G. Sapos

SAPOS®	HEPS	EPS	GPPS
Procedure	Real time	Real time	Postprocessing
Accuracy – position	0,01 – 0,02 m	0,3 – 0,8 m	≤ 0,01 m
Accuracy – height	0,02 – 0,03 m	0,5 – 1,5 m	0,01 – 0,02 m
Transmission technique	Ntrip over Internet (GPRS, UMTS, LTE), GSM	Ntrip over Internet (GPRS, UMTS, LTE)	Internet (Web Server)
Clocking unit	1 sec	1 sec	≥ 1 sec
Unit	1 min	inapplicable	1 min
SAPOS®-fee per unit	0,10 EUR	150,- EUR p. a.	0,20 EUR
Standard, Format	RTCM 3, RTCM 2.3	RTCM 2.3	RINEX 2.1

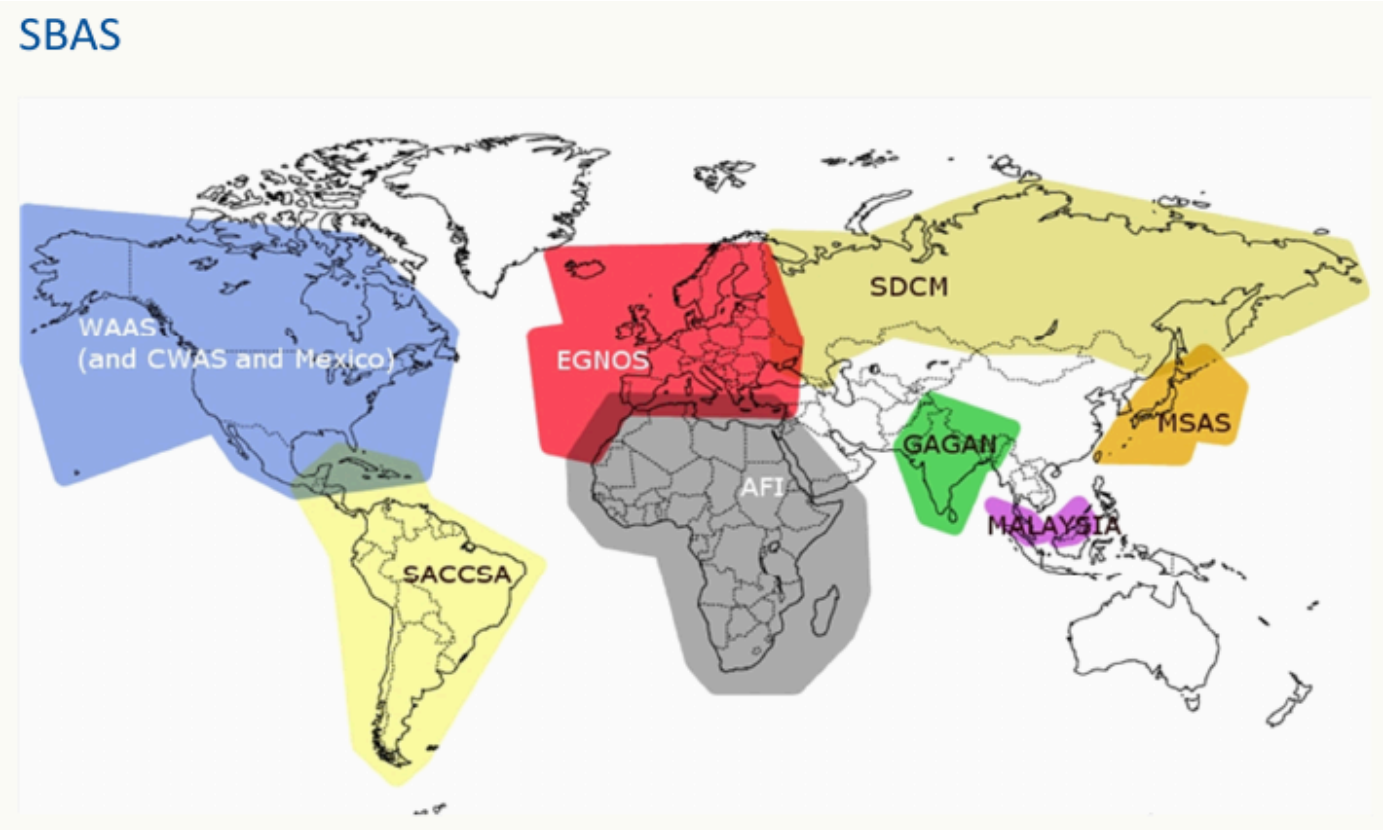
Augmentation Systems

- GNSS Augmentation Systems increase
  - Accuracy
  - Integrity
  - Robustness
  - Availability
- Important for safety critical applications (landing in aviation)
- 2 Types:
  - Ground based Augmentation Systems (GBAS)
  - Space based Augmentation Systems (SBAS)
- Delivery of
  - Correction data for satellite
  - Monitoring of the system functionality (monitoring)
- No info on multipath and troposphere

SBAS

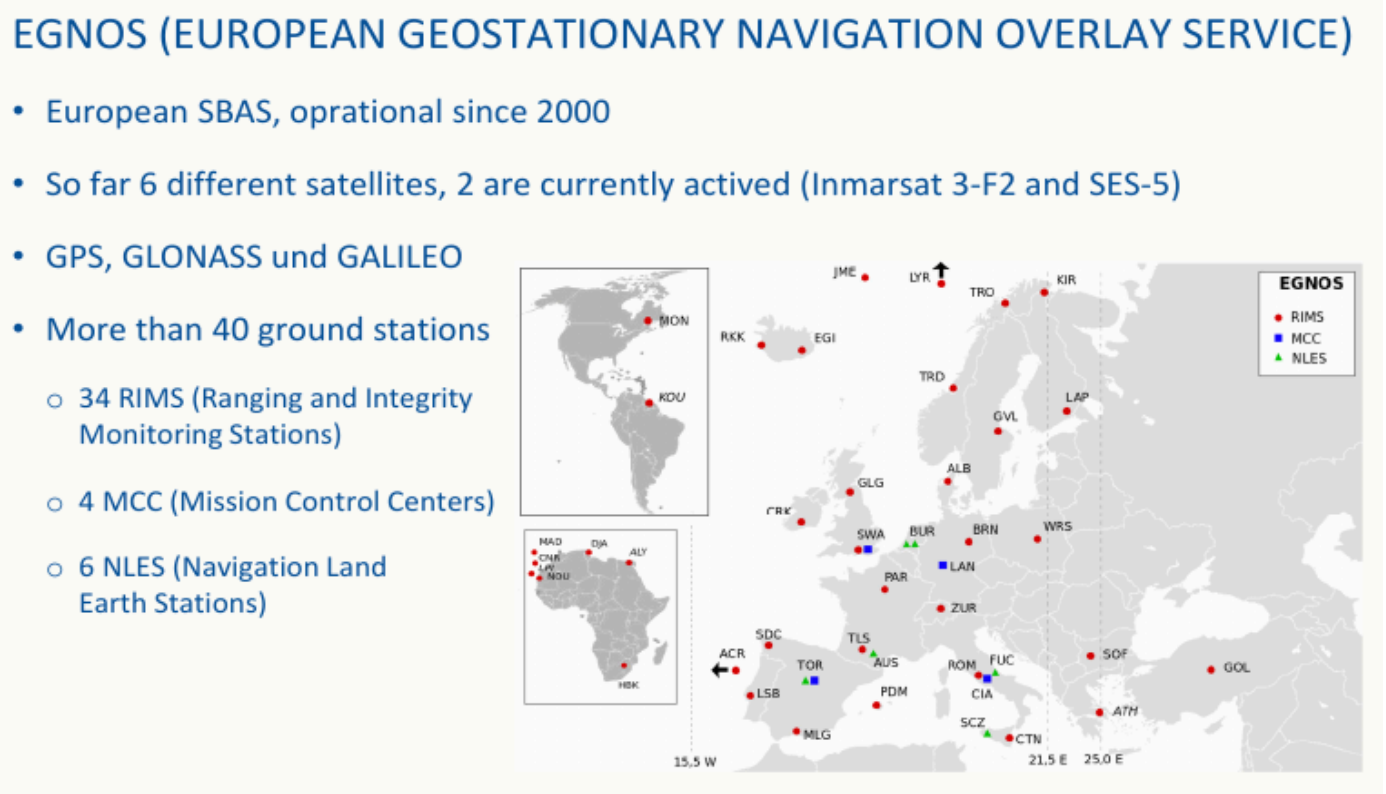
- At least two geostationary satellites
- E.g. WAAS, EGNOS, GAGAN, SDMC, ...
- Info
  - Correction data for accuracy
  - GNSS-like signals from SBAS satellites -> increasing availability, continuity, accuracy
  - Integrity info - 6sec notification after error

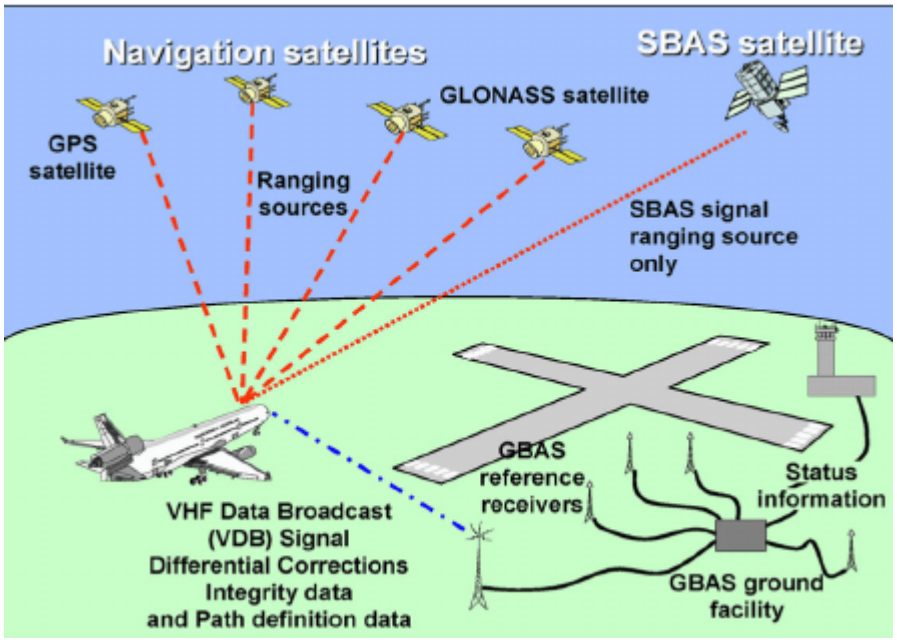
- Signals: received via normal reciever (no special radio link is necessary)



GBAS

- mainly around local areas (e.g. ariports)
- Examples: WAAS, pseudolites



- 
Navigation satellites
SBAS satellite
GLONASS satellite
GPS satellite
Ranging sources
SBAS signal ranging source only
GBAS reference receivers
Status information
VHF Data Broadcast (VDB) Signal
Differential Corrections
Integrity data
and Path definition data
GBAS ground facility